

MACHINEPULSE · DESIGN DECISIONS REGISTER

MachinePulse design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions from MPL-DDR-001 to 003 and the review note

MachinePulse design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Design for construction: accept the ten changes that make the pod buildable (separate boss on a set screw, tapped magnet holes, moved stand-offs with nylon screws, grommet, carrier board on stand-offs, three glands, corner pillars, bonded accelerometer, made probe clip, light pipe)	Accept; accept with changes; reject	Accept: none changes what the pod does, its pitch or its safety case	The whole build plan	MPL-DDR-003, Table 1
2	How the 5 V supply enters the box	(a) cut USB cable through an M12 gland to a screw terminal, as modelled; (b) IP67 panel-mount USB-C socket, about USD 5 more	(a) for the prototype	Power gland, carrier 5 V terminal (build plan 3.4.1, wire 1)	MPL-DDR-003, A1
3	Where the status button goes	(a) on the carrier board, pressed with the lid off, as modelled; (b) sealed button on the lid, about USD 1.50 more	(a) for the prototype	Carrier board layout; lid drilling	MPL-DDR-003, A2; review note 2026-09-26, item 1
4	How to treat R4 (energy per shift) with one current transformer	(a) accept at risk with the power factor curve (8.5 % RSS); (b) relax R4 to a relative energy trend; (c) add a voltage reference, about USD 10 more	None made; the earlier note offered (a) or (b) without choosing	None in the build; firmware and gateway at TRL 4	MPL-DDR-001, O2; MPL-DDR-002
5	Co-design partner for the alerts and dashboard	Who decides what "changed" should mean to an operator	None made	None in the build	MPL-DDR-001, O1
6	Enclosure colour in the renders	(a) two-tone lid and base, as rendered; (b) a single-colour stock box	Buy a single-colour box; keep two-tone as a later option	None in the prototype build (stock box)	Review note 2026-09-26, item 2

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The current transformer's 3.5 mm plug passes the M16 gland's seal	Otherwise the plug must be fitted after the lead is through, or a larger gland used	MPL-DDR-003, P6
2	The pot magnets' M6 studs are at least 5 mm long and can be trimmed	The stud and the set screw share the block's left hole	MPL-DDR-003, P1, P2
3	The silicone grommet fits a 24 mm hole in 2.5 mm panel and grips a 20 mm boss	It seals the floor and must stay soft so it carries little vibration or heat	MPL-DDR-003, P4
4	The stock box: four corner pillars about 7 mm across, 2.5 mm walls, continuous use at 70 °C	The gland heights, the carrier position and the thermal model assume them	MPL-DDR-003, P7; MPL-CAL-001, G
5	The controller board: about 22 x 51 mm, two 20-pin rows 19.4 mm apart, no octal PSRAM	The carrier's header sockets are spaced for it	MPL-DDR-003, P5; MPL-CAL-001, G4
6	The light pipe's hole size (3.2 mm clearance or 3.0 mm press fit) and its length (about 29 mm)	The lid hole and the LED position follow from it	MPL-DDR-003, P10
7	The probe clip's disc magnets are rated 150 °C or more	The clip sits on frames up to 100 °C	MPL-DDR-003, P9
8	The accelerometer breakout's size (about 18 x 18 mm) and that its back can be bonded flat	It is bonded to the 20 mm boss top	MPL-DDR-003, P8

■ Value engineering

Value-engineering target: USD 81 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 83.50 (USD 2.50 over the target). Main cost drivers and savings worth trying:

- The largest lines are the accelerometer (USD 15, the least certain price), the current transformer (USD 10), the controller (USD 9), the box with its glands (USD 9.50) and the power adapter (USD 8).
- Making the design constructable added USD 2.50: the third gland and the larger M16 gland (USD 0.50), the carrier board's sockets, terminals and light pipe (USD 1.00), the made probe clip with its magnets in place of a bought clip (USD 0.50 net) and the grommet, set screw and carrier fixings (USD 0.50).
- Savings worth trying: the steel pads (USD 4 for two) are needed only on aluminium-framed machines and could be left out of the standard kit; the accelerometer price should be checked against current breakout boards; an adapter and cable bought in quantity, or reused from an existing USB charger at the site, would cover the gap.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D10: Wi-Fi first, certified 5 V adapter, IIS3DWB-class sensor, one CT on one phase, first-week baseline alerts, TwinKit or any MQTT broker, stock ABS box on 5 mm nylon stand-offs, makerspace pilot, no budget change, features every minute and a spectrum every 15 minutes	Amish: "i accept all your recommendations, go with them across all repos."	MPL-DDR-001, MPL-DDR-002
2026-09-25	N1 high-temperature pot magnets rated 120 °C; N2 R8 restated (2 °C to 85 °C, 4 °C from 85 to 100 °C); N3 keep the R3 and R5 targets and measure first at TRL 4	Amish, same instruction	MPL-DDR-002
2026-09-30	Build plans are written in the approved format, and the design is made physically buildable as the pictures are drawn	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	MPL-DDR-003 (changes open for review, decision 1 above)
2026-09-30	Open decisions are kept in this register, not in the build plan	Amish: "don't log outstanding decisions in this build plan - that is not the place for it."	This register
2026-10-01	budget_usd is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens."	This register, Value engineering